



# Cloud-Based Pharmacy Management System

IT MANAGEMENT FINAL PROJECT · CBPMS-2026

- **Andrea Kitanovska**

Project Manager

- **Ana Milenkoska**

Frontend & Docs

- **Samoil Gjorgiev**

Backend Developer

- **Hristijan Kochovski**

Frontend & Docs

# The Problem

## Fragmented Records

Patient data stays local, not shared across branches.

## Paper Prescriptions

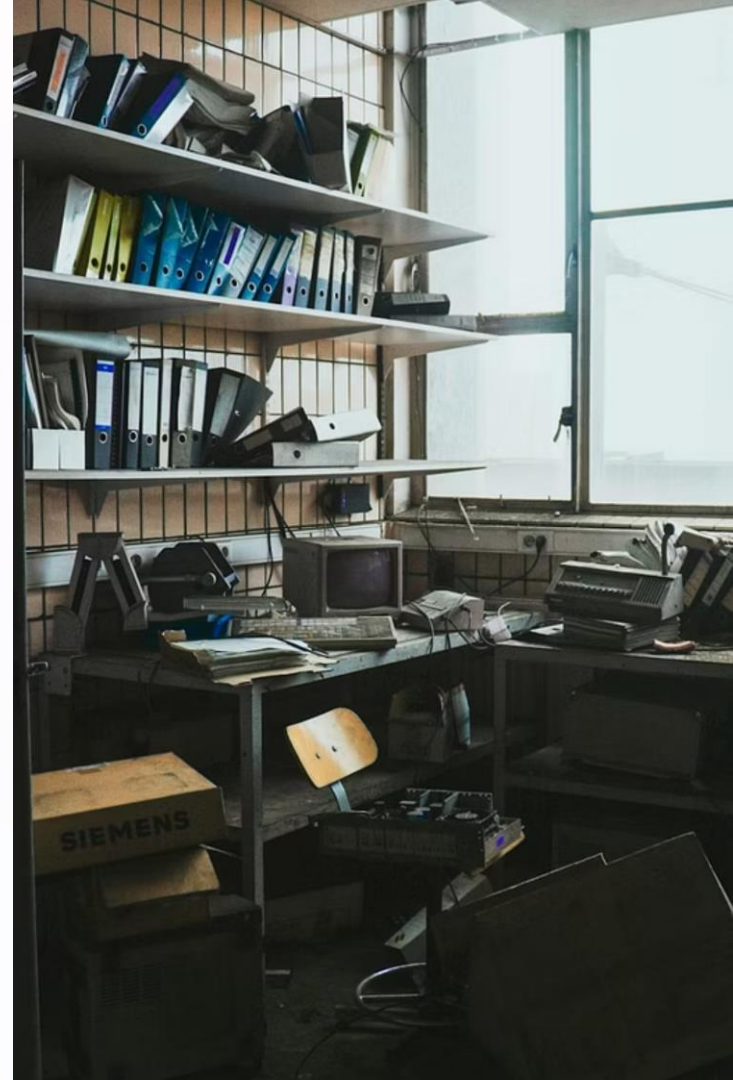
Paper scripts cause delays, errors and re-entry.

## Manual Stock Checks

Inventory is tracked by hand, so low stock and expiry dates are easy to miss.

## No Synchronization

Branches work in isolation and updates are not real time.



# Project Goal



The **Cloud-Based Pharmacy Management System** replaces fragmented tools with one centralized platform.

## → Centralized patient records

Access anywhere and anytime.

## → Automated stock & alerts

Real-time tracking with low-stock alerts.

## → Digital prescriptions

Issued and fulfilled in one system.

## → Role-based access

Secure login for doctors, pharmacists, and admins.

# Main Users



## Doctor

Fast access to patient history, allergies, and digital prescriptions.



## Pharmacist

Real-time prescriptions, stock updates, and alerts.



## Admin

Manages roles, permissions, and access control.

# Main Features



## Patient Search by Unique ID

Instant access to history, allergies, and prescriptions.



## Prescription Completion

Mark fulfilled in one step; stock updates automatically.



## Role-Based Access

Secure login with role-based permissions.



## Digital Prescriptions

Create and link prescriptions to patient records.



## Stock & Expiry Alerts

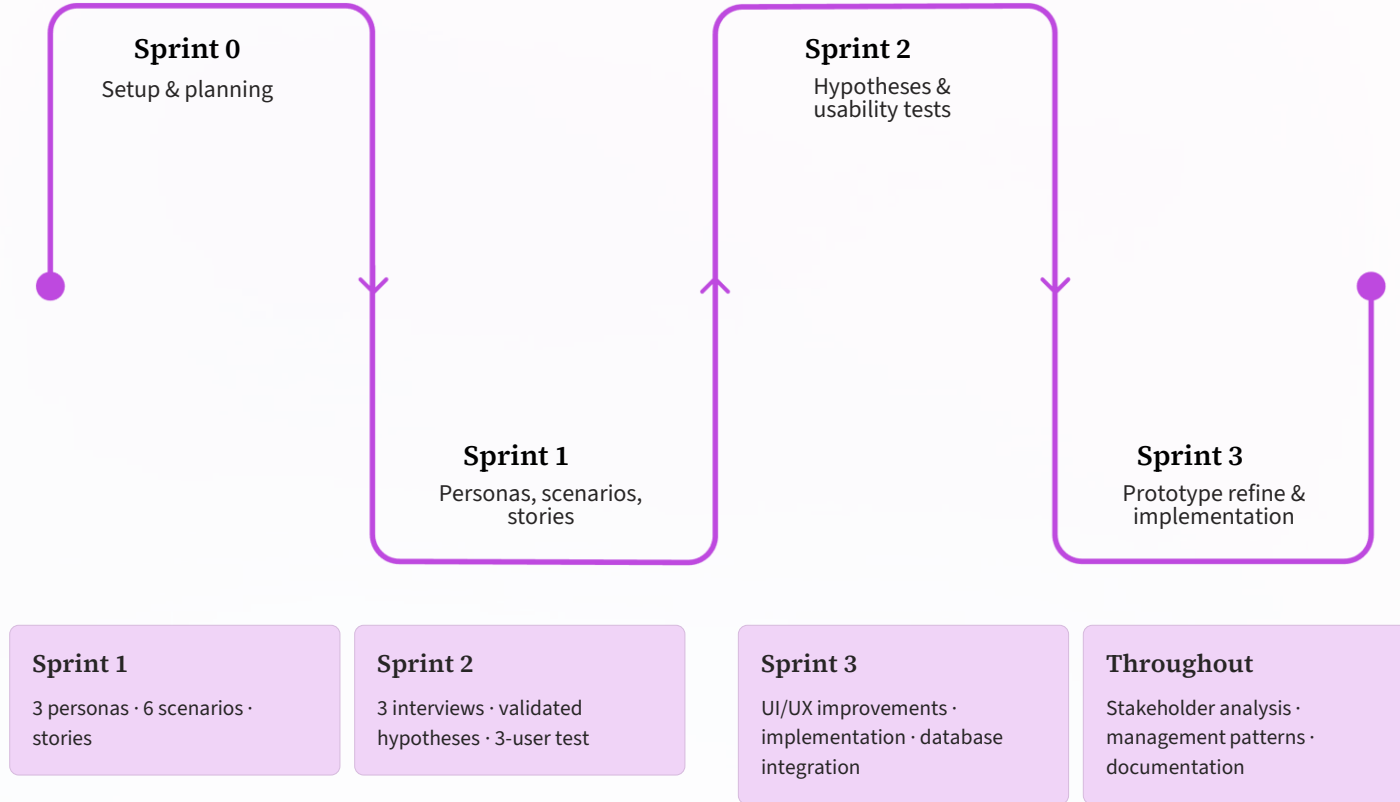
Alerts for low stock and near expiry.



## Real-Time Synchronization

Data stays consistent across branches.

# How the Project Developed



# Key Findings from Testing

- ✓ All 3 participants completed every task successfully. Core flows are clear.

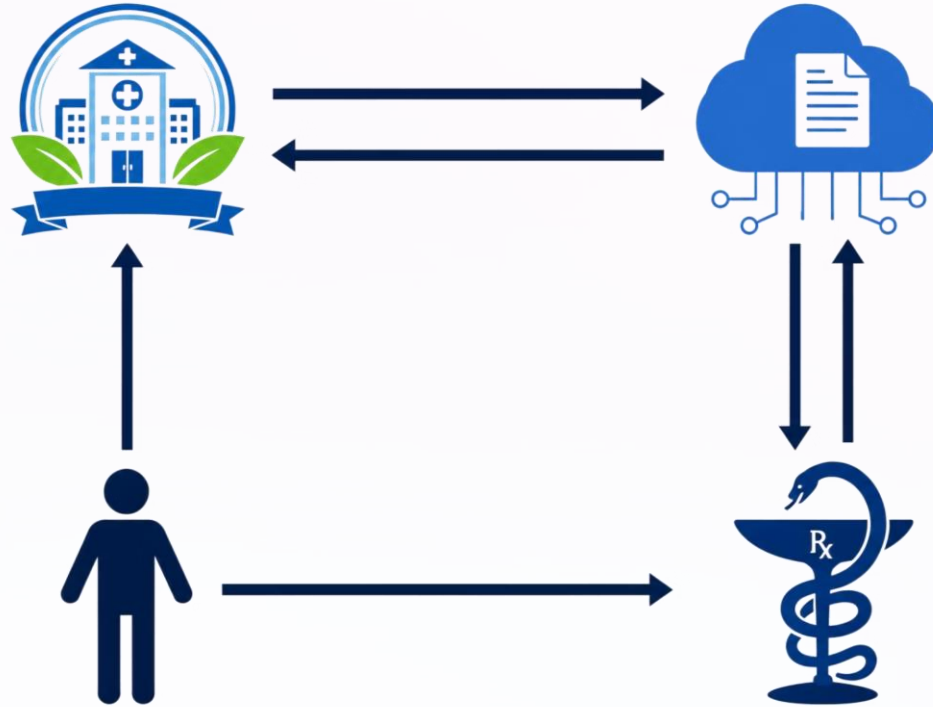
**Easiest task:** locating low-stock alerts.

**Most challenging:** finding a patient and completing a prescription in one flow.

## Improvements Identified

- Clearer confirmation messages
- Better Add Patient flow
- Simpler prescription completion
- Highlight allergies at the top
- Show last update and sync status

# System Demo Flow



# Conclusions & Takeaways



## 1 Healthcare complexity

Safety-critical decisions matter.

## 2 Usability & testing

Testing found issues specs missed.

## 3 Agile teamwork

Sprints kept feedback flowing.

## 4 Engineering management

Patterns shape project outcomes.